

Qns	Answer	Marks
7(a)(i)	The force on the electron is (always) perpendicular to the <u>velocity</u>/perpendicular to the <u>direction of travel/motion</u> of the electron. Thus no work is done by the force on the particle. (The speed remains unchanged and the force changes its direction.)	B1
7(a)(ii)	$F = Bqv$ $= (0.0050)(1.60 \times 10^{-19})(5.0 \times 10^6)$ $= 4.0 \times 10^{-15} \text{ N}$	C1 A1
7(a)(iii)	Magnetic force provides the centripetal force. $F = ma$ $= m \left(\frac{v^2}{r} \right)$ $r = \frac{mv^2}{F}$ $= \frac{(9.11 \times 10^{-31})(5.0 \times 10^6)^2}{4.0 \times 10^{-15}}$ $= 5.69 \times 10^{-3}$ $= 5.7 \text{ mm}$ $F = ma$ $(Bqv) = m \left(\frac{v^2}{r} \right)$ $r = \frac{mv}{Bq}$ $= \frac{(9.11 \times 10^{-31})(5.0 \times 10^6)}{(0.0050)(1.60 \times 10^{-19})}$ $= 5.69 \times 10^{-3}$ $= 5.7 \text{ mm}$	B1 B1 c
7(a)(iv)	Radius of path is proportional to its mass. Mass of proton is <u>many order of magnitude larger</u> than the electron implies the radius of the proton would be many orders of magnitude larger (and is too large as compared to a typical laboratory).	B1 B1
7(b)	$F = ma \Rightarrow eE = ma \Rightarrow e \frac{V}{d} = ma \Rightarrow a = \frac{eV}{md} \quad \dots\dots (1)$ $t = \frac{x}{v} \quad \dots\dots (2)$ Substituting (1) & (2): $y = vt + \frac{1}{2}at^2$ $= \frac{1}{2} \left(\frac{eV}{md} \right) \left(\frac{x}{v} \right)^2$ $= \left(\frac{eV}{2mdv^2} \right) x^2$	M1 M1 Correct substitution into the correct kinematics formula A1

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8(a)(i)1	α particles have greater ionisation power than X-rays, γ rays and	B1